

JavaScript DOM coding Lessons

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Simple HTML page with innerHTML update

```
<!DOCTYPE html>
```

```
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="output"></div>
  <div id="one"></div>
  <script src="/js/app1.js"></script>
</body>
</html>
```

```
console.dir(document);
console.dir(document.body);
console.log(document.URL);
console.log(document.title);
console.log(document.body.innerHTML);
document.body.innerHTML = '<h1>Laurence Svekis New</h1>';
```

Page Element Selection

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
```

```

    <div class="output"></div>
    <div id="one"></div>    <div class="output"></div>    <div
class="output"></div>
    <script src="/js/app1.js"></script>
</body>
</html>

```

```

//document.write('one');
//document.write('two');
const oneElement = document.getElementById('one');
oneElement.innerHTML = '<h1>Test</h1>';
oneElement.innerHTML = '<h1>New</h1>';

const output1 = document.getElementsByClassName('output');
console.log(output1);
for(let i=0;i<output1.length;i++){
    console.log(output1[i]);
    output1[i].innerHTML = `<h1>Index : ${i}</h1>`;
}

const divs = document.getElementsByTagName('div');
console.log(divs);

const el1 = document.querySelector('#one');
console.log(el1);
const el2 = document.querySelector('.output');
console.log(el2);
const el3 = document.querySelector('h1');
console.log(el3);

```

```

const el1s = document.querySelectorAll('#one');
console.log(el1s);
const el2s = document.querySelectorAll('.output');
console.log(el2s);
const el3s = document.querySelectorAll('div');
console.log(el3s);

for(let i=0;i<el3s.length;i++){
  el3s[i].innerHTML = `<h1>Div : ${i}</h1>`;
}

el3s.forEach((e,index)=>{
  e.innerHTML = `<h2>ELE ${index}</h2>`;
})

```

Select and update only divs with a class of output

```

<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1 class="output">Laurence Svekis</h1>
  <div class="output"></div>
  <div id="one"></div>
  <div class="output"></div>
  <div class="output"></div>

```

```
<script src="/js/app2.js"></script>
</body>
</html>
```

Laurence Svekis

Index : 0
Index : 1
Index : 2

```
const eles = document.querySelectorAll('div.output');
console.log(eles);
eles.forEach((el,ind) =>{
  el.innerHTML = `Index : ${ind}`;
})
```

Create Elements and Remove Elements

Laurence Svekis

Test 1

Index 0

Index 1

Index 2

```
<!DOCTYPE html>
<html>
```

```
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="output"></div>
  <script src="/js/app3.js"></script>
</body>
</html>
```

```
const output1 = document.querySelector('.output');
const output2 = document.createElement('div');
console.log(output1);
console.log(output2);
```

```
output1.innerHTML = 'Test 1';
output2.innerHTML = 'Test 2';
```

```
document.body.append(output2);
const res1 = output1.append(output2);
document.body.prepend(output2);
```

```
const res2 = output1.appendChild(output2);
console.log(res1);
console.log(res2);
```

```
output1.removeChild(output2);
console.log(output2);
output1.append(output2);
```

```

console.log(output2.parentNode);
output2.parentNode.removeChild(output2);

for(let i=0;i<10;i++){
  const ele = document.createElement('div');
  ele.innerHTML = `Index ${i}`;
  output1.append(ele);
}
/*
let html = "";
for(let i=0;i<10;i++){
  html += `<div>HTML ${i}</div>`;
}
console.log(html);
output1.innerHTML = html;
*/

```

Page Element Style Property

Laurence Svekis



```

const main = document.querySelector('.output');

```

```

for(let i = 0;i<10;i++){
  const ele = document.createElement('div');
  ele.textContent = `Ele ${i}`;
  ele.classList.add('box');
  ele.style.display = 'inline-block';
  ele.style.border = '1px solid #eee';
  ele.style.width = '100px';
  ele.style.fontFamily = 'Arial';
  main.append(ele);
}

const eles = document.querySelectorAll('.box');
console.log(eles);

eles[0].style.color = 'red';
eles[1].style.border = '1px solid #ddd';
eles[2].style.display = 'none';
eles[3].style.padding = '10px';
eles[4].style.margin = '10px';
eles[5].style.width = '200px';
eles[6].style.textAlign = 'right';
eles[7].style.backgroundColor = 'yellow';

```

Interactive Page elements Events

```

const ele1 = document.querySelector('h1');
const ele2 = document.querySelector('.output');

```



```

document.body.onload = ()=>{
  console.log('ready');
}
let counter = 0;
ele1.onclick = ()=>{
  adder('clicked');
};
ele1.onmousedown = ()=>{
  adder('mouse down');
}
ele1.onmouseup = ()=>{
  adder('mouse up');
}
ele1.onmouseover = ()=>{
  adder('mouse over');
}
ele1.onmouseout = ()=>{
  adder('mouse out');
}
ele2.onclick = output;

function adder(val){
  counter++;
  console.log(val);
  ele2.textContent = `${val} click ${counter}`;
}

function output(){
  counter++;

```

```
ele2.textContent = `click ${counter}`;  
}
```

Event Object values

```
const ele3 = document.querySelector('h2');  
  
ele3.onclick = adder1;  
ele3.onmousedown = adder1;  
ele3.onmouseup = adder1;  
function adder1(e){  
  counter++;  
  console.log(e.type);  
  e.target.textContent = `Event Type : ${e.type} ${counter}`;  
}
```

Event Listeners vs Event Handlers

click 2	<u>app5.js:7</u>
click 3	<u>app5.js:11</u>
click 4	<u>app5.js:14</u>

Use the event listeners when in doubt, event listeners are not overwritten and there is more control over the event.

```
const h2 = document.querySelector('h2');  
  
h2.onclick = ()=>{
```

```
    console.log('click 1');
  }
  h2.onclick = ()=>{
    console.log('click 2');
  }

  h2.addEventListener('click',()=>{
    console.log('click 3');
  })
  h2.addEventListener('click',()=>{
    console.log('click 4');
  })
```

UseCapture Event Listener Optional

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
  <style>
    .output {
      border:1px solid black;
      width:100px;
      height:50px;
      text-align:center;
      background-color:white;
    }
  </style>
</head>
```

```

<body>
  <h1>Laurence Svekis</h1>
  <h2>Click Me</h2>
  <div class="output"></div>
  <script src="/js/app6.js"></script>
</body>
</html>

```

UseCapture Example flows up from the child to parents or true to flow from parent to children.

```

const el1 = document.querySelector('h2');
const el2 = document.querySelector('.output');
const el3 = document.querySelector('h1');
el2.textContent = 'Click';
let temp = document.querySelector('.output');
for(let i=0;i<5;i++){
  const ele = document.createElement('div');
  ele.classList.add('output');
  ele.textContent = ` ${i+1} Element `;
  ele.id = ` Ele${i} `;
  ele.style.height = 300 - i*50 + 'px';
  ele.style.width = 100 + i*50 + 'px';
  temp.append(ele);
  ele.addEventListener('click',()=>{
    console.log(i);
  });
  temp = ele;
}

el1.addEventListener('click',output);

```

```
//el2.addEventListener('click',output);  
el3.addEventListener('click',output);
```

```
function output(e){  
  console.log(e.target.id);  
  //console.log(e.type);  
}
```

```
/*  
h2.addEventListener('click',(e)=>{  
  console.log(e.type);  
});  
*/
```

Mouse Movement Events

Leave	Leave
Leave	Leave
Leave	

Laurence Svekis

Click 1

Click 2

mouseleave 26

```
const btns = document.querySelectorAll('button');
const main = document.querySelector('.output');
const main1 = document.createElement('div');
document.body.prepend(main1);
```

```
for(let i=0;i<5;i++){
  const ele = document.createElement('div');
  ele.style.width = '100px';
  ele.style.display = 'inline-block';
  ele.style.border = '1px solid black';
  ele.textContent = `Element ${i}`;
  ele.addEventListener('mouseenter',()=>{
    ele.style.color = 'red';
    ele.textContent = 'Enter';
```

```

    })
    ele.addEventListener('mouseleave',()=>{
        ele.style.color = 'black';
        ele.textContent = 'Leave';
    })
    ele.addEventListener('mouseover',()=>{
        ele.textContent = 'Over';
    })
    ele.addEventListener('mouseout',()=>{
        ele.textContent = 'Out';
    })
    main1.append(ele);
}

```

```

let counter = 0;
btns[0].addEventListener('click',output);
btns[1].addEventListener('click',output);
btns[0].addEventListener('mousedown',show1);
btns[0].addEventListener('mouseup',show1);
btns[0].addEventListener('mouseenter',show1);
btns[0].addEventListener('mouseleave',show1);

```

```

function show1(e){
    counter++;
    main.textContent = `${e.type} ${counter}`;
}

```

```
function output(e){  
  counter++;  
  e.target.removeEventListener('click',output);  
  main.textContent = `${e.type} ${counter}`;  
}
```

Form Field Event listeners

Laurence Svekis

d
test

```
const myInput = document.querySelector('input');  
const output = document.querySelector('.output');  
for(let i=0;i<5;i++){  
  const ele = document.createElement('input');  
  output.append(ele);  
  ele.addEventListener('focus',()=>{
```



```
        ele.style.backgroundColor = 'red';
    })
    ele.addEventListener('focusout',()=>{
        ele.style.backgroundColor = 'white';
    })
    ele.onChange = ()=>{
        console.log(ele.value);
    }
    ele.addEventListener('keydown',add);
    ele.addEventListener('keyup',add);
}

function add(e){
    console.log(e.type);
    console.log(e.key);
}

myInput.onfocus = ()=>{
    console.log('focused');
}
```

List Exercise on events

Laurence Svekis

- Index 0
- Index 1
- Index 2
- Index 3

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="output"></div>
  <input type="number">
  <script src="/js/app9.js"></script>
</body>
</html>
```

```
const myVal = document.querySelector('input');
const main = document.querySelector('.output');
```

```

myVal.addEventListener('change',()=>{
  main.innerHTML = "";
  const ul = document.createElement('ul');
  main.append(ul);
  if(!isNaN(myVal.value)){
    for(let i=0;i<myVal.value;i++){
      const li = document.createElement('li');
      li.textContent = `Index ${i}`;
      ul.append(li);
    }
  }
})

```

Window Object WOM

```

console.dir(window);
const main = document.querySelector('.output');
main.style.height = '2000px';
window.addEventListener('resize',output);
window.addEventListener('scroll',()=>{
  console.log(window.scrollY);
})

```

```

function output(e){
  console.log(window.innerWidth);
}

```

DOM element attributes

Using the setAttribute, element attributes can be added. Elements like input fields, images and hyperlinks commonly have attributes.

Example of elements with attributes added by JavaScript code

```

<a href="http://www.google.com" target="_blank" class="box box1 box2">Click
me</a>
</div>
<input type="number" min="1" max="10" required>
```

```
const main = document.querySelector('.output');
```

```
const myInput = document.querySelector('input');
myInput.setAttribute('type','number');
myInput.setAttribute('min','1');
myInput.setAttribute('max','10');
myInput.setAttribute('required','');
myInput.value = '5';
```

```
const img1 = document.createElement('img');
main.append(img1);
img1.setAttribute('src','//via.placeholder.com/150');
img1.setAttribute('id','img2');
```

```
img1.id = 'img1';
console.log(img1.src);
```

```
const hyperL = document.createElement('a');
hyperL.textContent = 'Click me';
hyperL.setAttribute('href','http://www.google.com');
hyperL.setAttribute('target','_blank');
```

```
hyperL.setAttribute('class','box box1 box2');  
main.append(hyperL);
```

Element Get and Has attribute

update all the page hyperlink elements to have a target of `_blank`.

```
const a1 = document.querySelector('a');  
console.log(a1.hasAttribute('href'));  
console.log(a1.getAttribute('href'));  
  
const hypers = document.querySelectorAll('a');  
hypers.forEach(ele =>{  
  if(ele.hasAttribute('target')){  
    if(ele.getAttribute('target') != '_blank'){  
      ele.setAttribute('target','_blank');  
    }  
  }else{  
    ele.setAttribute('target','_blank');  
  }  
})
```

Adding and Removing Classes

Laurence Svekis

Element 0

Element 1

Element 2

Blue 3

Element 4

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
  <style>
    .red{
      color:red;
    }
    .blue{
      color:blue;
    }
  </style>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="output"></div>
  <script src="/js/app14.js"></script>
```

```
</body>
</html>
```

```
const h1 = document.querySelector('h1');
const main = document.querySelector('.output');

for(let i=0;i<10;i++){
  const ele = document.createElement('div');
  main.append(ele);
  ele.textContent = `Element ${i}`;
  ele.onclick = ()=>{
    console.log(ele);
    if(ele.classList.contains('blue')){
      ele.textContent = `Blue ${i}`;
      ele.classList.remove('blue');
    }else{
      ele.textContent = `Element ${i}`;
      ele.classList.add('blue');
    }
  }
}

h1.onclick = ()=>{
  console.log(h1.classList.contains('red'));
  h1.classList.toggle('red');
}
```

Traversing between elements

Laurence Svekis

- Item 1
- Item 2
- Item 3
- Item 4

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="main"></div>
  <script src="/js/app15.js"></script>
</body>
</html>
```

```
const main = document.querySelector('.main');
const ul = document.createElement('ul');
main.append(ul);
for(let i=0;i<9;i++){
  const li = document.createElement('li');
  li.textContent = `Item ${i+1}`;
  li.onclick = ()=>{
```



```

const par = li.parentNode;
console.log('First' + par.firstChild.textContent);
console.log('Last' + par.lastChild.textContent);
console.log('CUR' + li.textContent);
if(li.nextSibling !== null){
  console.log('NEXT' + li.nextSibling.textContent);
}
if(li.previousSibling !== null){
  console.log('PREV' + li.previousSibling.textContent);
}
}
ul.append(li);
}

console.log(ul.childNodes);
console.log(ul.childElementCount);
console.log(ul.children);

ul.childNodes.forEach(el=>{
  console.log(el.textContent);
})

```

List Exercise

Laurence Svekis

List Item 1	<button>Remove</button>	<button>Edit</button>
List Item 2	<button>Remove</button>	<button>Save</button>
List Item 3	<button>Remove</button>	<button>Edit</button>
List Item 4 test	<button>Remove</button>	<button>Save</button>
List Item 6	<button>Remove</button>	<button>Edit</button>
test	<button>Remove</button>	<button>Edit</button>

Add New

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
  <style>
    .striker {
      text-decoration:line-through;
      color:red;
      cursor:grab;
    }
    .btns{
      margin:0 5px;
      font-size:0.7em;
      color:white;
    }
    ul, li{
      margin:0;
```

```

padding:0;
list-style-type:none;
}
.content{
font-size:1.2em;
padding:3px;
width:50%;
display:inline-block;
}
</style>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="main"></div>
  <script src="/js/app16.js"></script>
</body>
</html>

```

```

const main = document.querySelector('.main');
const ul = adder('ul',main,'ul'," ");
const inEle = adder('input',main,'input'," ");
const btn = adder('button',main,'btn','Add New ');
inEle.setAttribute('type','text');
inEle.setAttribute('placeholder','List Item');

btn.addEventListener('click',newListItem);

for(let i=0;i<6;i++){
  inEle.value = `List Item ${i+1}`;
  newListItem();
}

```

```
}
```

```
function newListItem(){
  const li = adder('li',ul,'li');
  const ele = adder('span',li,'content',inEle.value);
  const delEle = adder('button',li,'btns','Remove');
  const ediEle = adder('button',li,'btns','Edit');
  ele.setAttribute('contenteditable','false');
  delEle.style.backgroundColor = 'red';
  ediEle.style.backgroundColor = 'green';
  ele.onclick = ()=>{
    if(ele.getAttribute('contenteditable') == 'false'){
      ele.classList.toggle('striker');
    }
  }
  delEle.onclick = ()=>{
    //console.log(delEle.parentNode.firstChild);
    li.remove();
  }
  ediEle.onclick = ()=>{
    if(ele.getAttribute('contenteditable') == 'true'){
      ele.setAttribute('contenteditable','false');
      li.style.backgroundColor = '#fff';
      ediEle.textContent = 'Edit';
      ediEle.style.backgroundColor = 'green';
    }else{
      ele.setAttribute('contenteditable','true');
      ele.classList.remove('striker');
    }
  }
}
```

```

        li.style.backgroundColor = '#ddd';
        ediEle.textContent = 'Save';
        ediEle.style.backgroundColor = 'blue';
    }
}
inEle.value = "";
}

```

```

function adder(t,parent,c,tc){
    const ul = document.createElement(t);
    ul.classList.add(c);
    ul.textContent = tc;
    return parent.appendChild(ul);
}

```

requestAnimationFrame movement

Laurence Svekis



```

<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
  <style>
    .main{
      position:absolute;
      width:100px;
      height:100px;
      left:0;
      top:0;
      background-color:red;
    }
  </style>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="main">Move</div>
  <script src="/js/app17.js"></script>
</body>
</html>

const main = document.querySelector('.main');
const pos = {x:0,y:0};
//const intVal = setInterval(updater,100);
let intAni = requestAnimationFrame(moveEle);

function moveEle(){
  pos.x+=10;
  pos.y+=5;

```

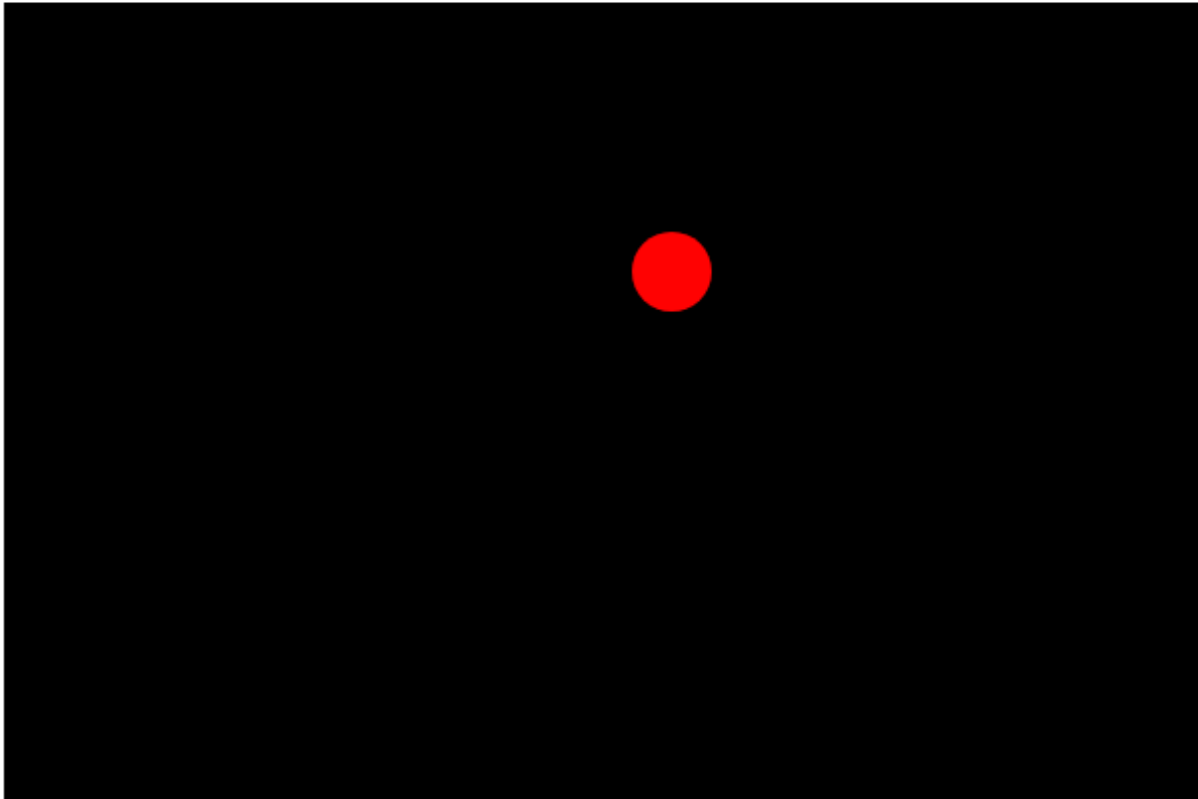
```
mover();  
if(pos.x > 200){  
    cancelAnimationFrame(intAni);  
}else{  
    intAni = requestAnimationFrame(moveEle);  
}  
}
```

```
function updater(){  
    pos.x+=10;  
    pos.y+=5;  
    if(pos.x > 200){  
        clearInterval(intVal);  
    }  
    mover();  
}
```

```
function mover(){  
    main.style.left = pos.x+'px';  
    main.style.top = pos.y+'px';  
}
```

Bouncing Ball Animation

Laurence Svekis



```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript DOM</title>
</head>
<body>
  <h1>Laurence Svekis</h1>
  <div class="main"></div>
  <script src="/js/app18.js"></script>
</body>
</html>
```



```

const main = document.querySelector('.main');
const h1 = document.querySelector('h1');
main.style.width = '600px';
main.style.height = '400px';
main.style.backgroundColor = 'black';

const ball = document.createElement('div');
const b = {x:50,y:30,w:40,h:40,dx:1,dy:1,speed:5,ani:{},move:false}
ball.style.backgroundColor = 'red';
ball.style.borderRadius = '50%';
ball.style.width = `${b.w}px`
ball.style.height = `${b.h}px`
ball.style.position = 'relative';
ball.style.left = `${b.x}px`;
ball.style.top = `${b.y}px`;
main.append(ball);

h1.addEventListener('click',()=>{
  if(!b.move){
    b.ani = requestAnimationFrame(mover);
    b.move = true;
  }else{
    cancelAnimationFrame(b.ani);
    b.move = false;
  }
})

function mover(){

```

```
if(b.x>600-b.w || b.x < 0){  
    b.dx *= -1;  
}  
if(b.y>400-b.h || b.y < 0 ){  
    b.dy *= -1;  
}  
b.x += b.dx * b.speed;  
b.y += b.dy * b.speed;  
ball.style.left = `${b.x}px`;  
ball.style.top = `${b.y}px`;  
if(b.move){  
    b.ani = requestAnimationFrame(mover);  
}  
}
```